Backoffice Workflow & Production Design

System Architecture & Implementation Record

1. System Overview

This document describes the complete production workflow for the optical backoffice system including power processing, job cards, lab routing, allocation, and billing. The Backoffice module is responsible for: • Validating order lines • Managing R/L workflow • Allocation and pricing • Job card generation • Routing to labs/vendors

2. Folder Structure

modules/ ■■■ backoffice_management.py ■■■ power_engine.py ■■■ batch_manager.py ■■■ sql_adapter.py ■■■ workflow/ ■ ■■■ engine.py ■ ■■■ status.py

3. Data Flow Architecture

Retail Punching → Database → Backoffice → Workflow Engine → Documents Flow: 1. Retail Punching saves order + metadata 2. SQL Adapter loads order 3. Backoffice validates and edits 4. Workflow engine runs 5. Documents generated

4. Order Line Schema

Each order line contains: Core: • product_id • product_name • brand • main_group • eye_side • billing_qty Power: • sph, cyl, axis, add_power • sph_out, cyl_out, axis_out Lens Parameters: • lens_type (from product master) • diameter • frame_type • fitting_height • frame_size • corridor • base_curve Workflow: • batch_allocation • batch_status • manufacturing_route • billing_total

5. SQL Adapter Integration

Main SQL Adapter Functions: • fetch_backoffice_orders() • fetch_full_order(order_no) • read_product_master() • fetch_last_product_price() Tables: • orders • order_lines • product_master • batch_stock • pricing_history

6. Power Engine

power_engine.py Functions: • vertex_correct_spherical() • vertex_correct_toric() Usage: Called from update_manufacturing_power() to calculate sph_out, cyl_out, axis_out.

7. Workflow Engine

workflow/engine.py Main functions: trigger_complete_workflow(line): • update_manufacturing_power() • update_batch_allocation() • update_line_billing() route_manufacturing(line): • INTERNAL_LAB • EXTERNAL_LAB • VENDOR • STOCK_VENDOR

8. Manufacturing Routing Logic

Routing Rules: Contact Lens → EXTERNAL_LAB Ophthalmic Lens: Internal capable → INTERNAL_LAB Else → VENDOR Other Products → STOCK_VENDOR Stored in: line['manufacturing_route']

9. Job Card Generation

Generated in backoffice_management.py Includes: • RX Power • Manufacturing Power • Lens Parameters • Base Curve • Frame Details • Batch Info Used for: • Internal Lab • External Vendor

10. Allocation & Billing

Allocation: • FIFO batch allocation • Power matched • Temp state handling Billing: • Max historical fallback price • Discount handling • Locked billing on save

11. Status Engine

workflow/status.py Defines: • PENDING • ALLOCATED • LAB_ORDER • IN_PRODUCTION • READY • DISPATCHED • DELIVERED Updated via: show_status_update_modal()

12. UI Integration Plan

Retail Punching UI: • Dropdowns for diameter, frame_type, fitting_height • Lens type from product master Backoffice UI: • R/L Processing • Allocation Editor • Power Edit • Pricing Panel • Job Card View

13. Future Enhancements

Planned: • Vendor API integration • PO automation • SLA tracking • Inventory forecasting • Lab workload balancing

Prepared for Production Deployment

Backoffice Optical Management System

Generated via ChatGPT Assistant